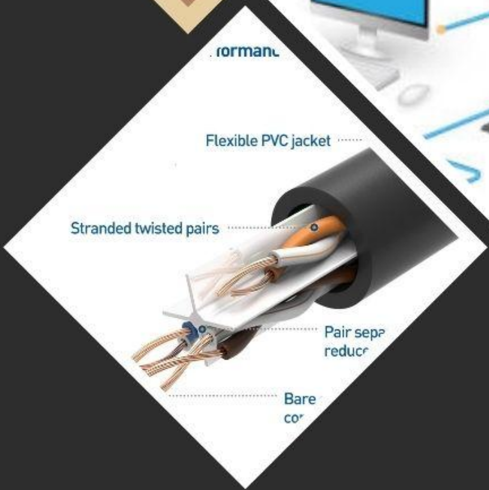

IE2004 - COMPUTER NETWORKS

ENTERPRISE NETWORK INFRASTRUCTURE REPORT

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Table of Contents

Section I – Infrastructure Design

Introduction

This report documents the comprehensive design of a network infrastructure for a small enterprise, aiming to provide robust wired and wireless connectivity across two primary functional zones: the Admin and Sales Department, and the IT and Meeting Rooms. The assignment follows a structured approach, encompassing logical and physical network design, detailed equipment and cabling specifications, and a full cost estimate, as required for small office or home office environments.

The logical network design, illustrated in the accompanying schematic, adopts a hierarchical star topology. Each department is equipped with a dedicated access switch, connecting local devices such as desktop PCs, laptops, printers, and wireless access points. These departmental switches are centrally linked to a core switch, which aggregates traffic and connects to an ISP router for secure and reliable internet access. Wireless coverage is ensured through strategically placed access points in both departments, supporting mobile and flexible work requirements.

The physical network design extends this logical framework, specifying the placement of cabling, patch panels, equipment racks, and power supplies to ensure efficient operation and ease of maintenance. Equipment specifications are provided for all active components, including switches, routers, and access points, with attention to standards compliance and future scalability. The cost estimate section details all required materials and equipment, supporting informed decision-making and budget planning.

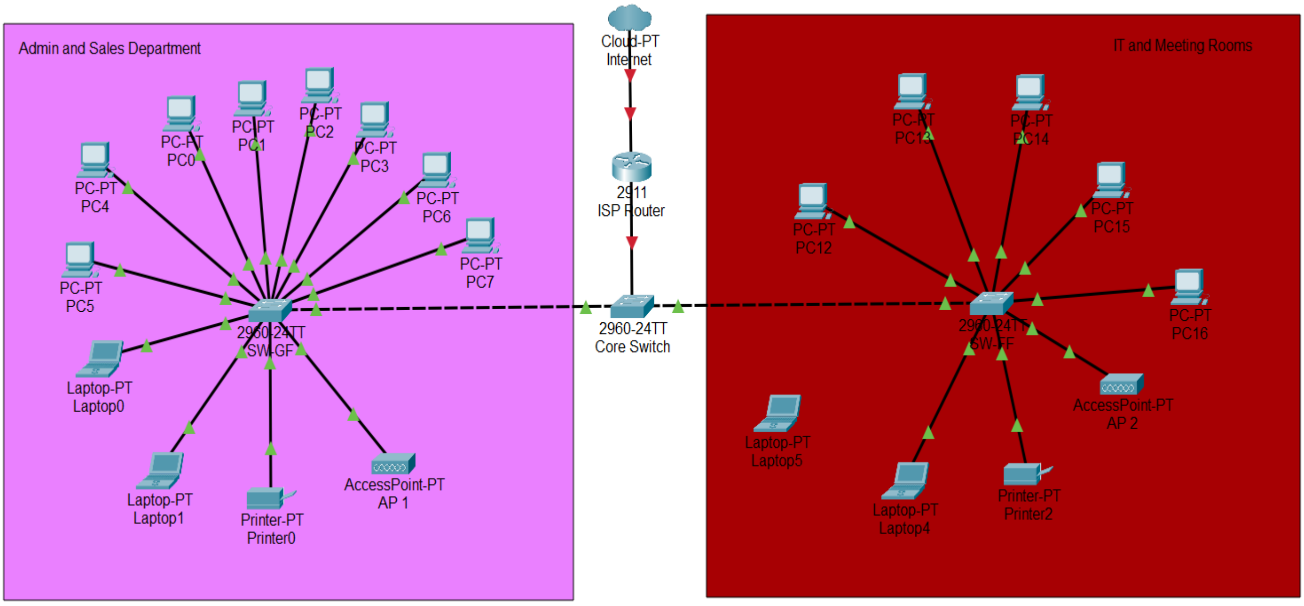
By following established best practices in network design-such as modular planning, scalability, manageability, and security-this report aims to deliver a network solution that meets current organizational needs while accommodating future growth and technological advancements.

Group Members and Their Roles

- IT23737068: Responsible for the logical network design, including the development of the network schematic and topology selection.
- IT23739598: Prepared the physical network design, including floor plan layout, equipment placement, and cabling routes
- IT23690370: Compiled equipment specifications and coordinated the preparation of the cost estimate.

Logical Network Design

This section outlines the logical network design, illustrating how all devices-including PCs, laptops, printers, switches, access points, and the router-are interconnected to provide efficient wired and wireless connectivity across the organization.



Connection Table:

Connection	Speed
Internet Cloud -> ISP Router	1 Gbps
ISP Router -> Core Switch	1 Gbps
Core Switch -> Access Switch – GF	1 Gbps
Core Switch -> Access Switch – FF	1 Gbps
Access Switch -> Access Points	1 Gbps
Access Switch -> PCs/Laptops/Printers	100 Mbps – 1 Gbps

Physical Network Design

The physical network design for this project spans two floors, each with a tailored layout to support the operational needs of the respective departments. The design ensures efficient connectivity, organized cabling, and ease of future expansion.

First Floor: Sales Department, Admin Office, and Server Room

- Sales Department:
  The Sales Department is situated on the first floor, featuring multiple workstations connected to a dedicated switch. This switch is placed in an equipment rack located within a separate room, streamlining cable management and maintenance. Wired connections are provided for all fixed devices, while an access point ensures wireless connectivity for mobile devices.
- Admin Office:
  The Admin Office, also on the first floor, is equipped with workstations connected to the departmental switch in the equipment rack. This setup enables secure and efficient access to administrative resources.
- Server Room:
  The Server Room on the first floor serves as the central hub for network infrastructure. It houses the core switch, which connects to all departmental switches across both floors. The ISP router is also installed here, providing Internet access to the entire network.

Second Floor: IT Department and Conference Room

- IT Department:
  The IT Department is located on the second floor and features a cluster of workstations connected via a dedicated switch. This switch is housed in an equipment rack placed within the IT Department for convenience. All workstations are connected to the rack using structured cabling, and a network printer is also integrated into this setup. The department is equipped with both wired and wireless connectivity, with an access point strategically positioned to ensure comprehensive Wi-Fi coverage.
- Conference Room:
  Adjacent to the IT Department, the conference room is equipped with network ports and wireless coverage. Devices in this room connect to the network through the same switch in the IT Department's equipment rack, allowing seamless access to shared resources during meetings.

Equipment Rack Specifications:

- Type: Wall-mounted racks
- Height: 12U
- UTP Patch Panel: 24-port patch panels in each rack to accommodate all endpoints on the respective floors
- Equipment Housed: Network switches, patch panels, cable management hardware
- Location: IT Department (second floor), Dedicated room for this (first floor)

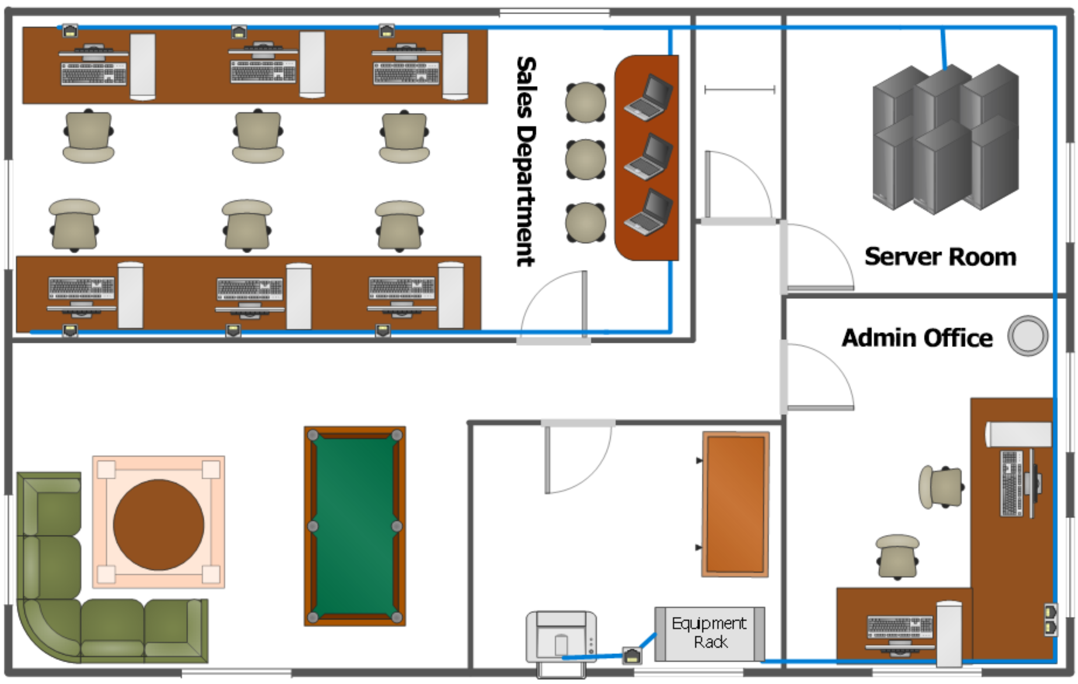
Cabling and Connectivity

- Cat 6 UTP cabling is used throughout, with cable paths clearly mapped to maintain lengths within the recommended 90-meter maximum between patch panels and wall outlets.
- All cables terminate at patch panels mounted in the equipment racks.
- Wireless access points are installed to the ceiling of both floors, with non-overlapping channels to minimize interference.

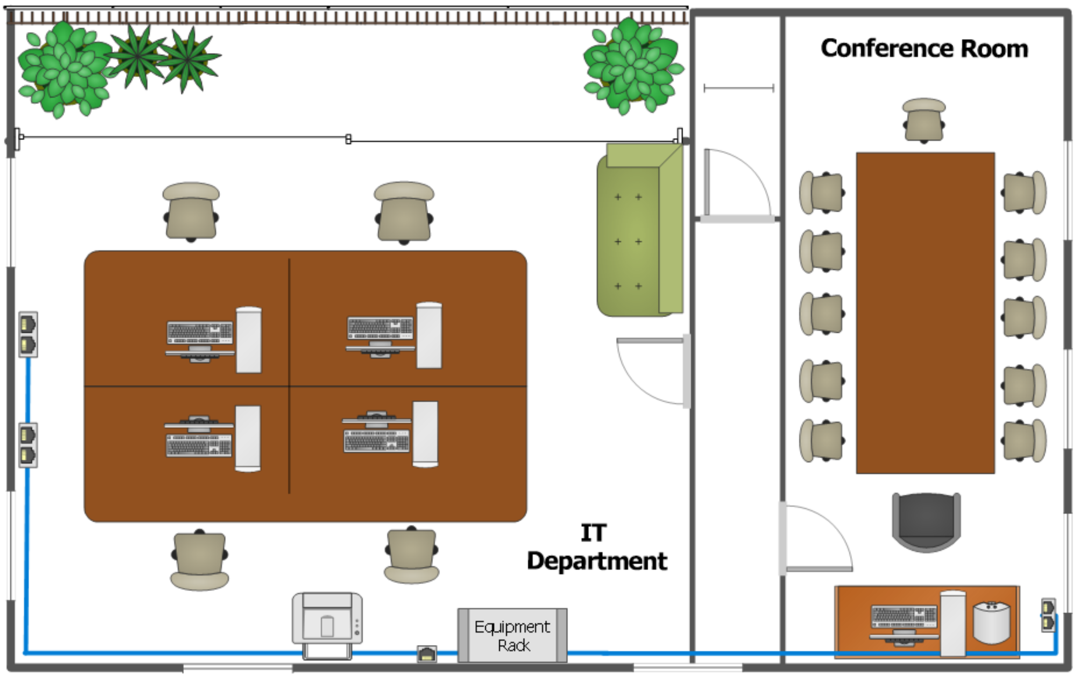
Power Supply:
All network equipment in both racks is powered via Uninterruptible Power Supplies (UPS) with the following specifications:

- 1500VA capacity for each rack, providing approximately 20-30 minutes of backup time at full load
- Automatic voltage regulation to protect against power fluctuations
- Network management capability for remote monitoring and controlled shutdown during extended outages

First Floor:



Second Floor:



Equipment Specifications

1. Switches

Specification	Value
Port Type	Copper (RJ-45) 10/100/1000 Mbps Ethernet
Port Count	24 ports per switch
Uplink Ports	2 × SFP fiber uplink ports (1 Gbps each)
PoE Requirements	PoE+ (IEEE 802.3at) on all copper ports
Speed per Port	1 Gbps (copper), 1 Gbps (fiber uplinks)
Backplane Capacity	Minimum 56 Gbps
Protocol Compliance	IEEE 802.3i, 802.3u, 802.3ab, 802.3z, 802.3af, 802.3at, 802.1Q, 802.1p, 802.1D, 802.1w, 802.1s, 802.1x, SNMP, RMON, LLDP, IGMP Snooping, Spanning Tree Protocols (STP/RSTP/MSTP)

2. Routers

Specification	Value
Port Type / Count	2 × Gigabit Ethernet (RJ-45), 1 × SFP (fiber, 1 Gbps), 1 × Console, 2 × USB
Packet Handling	Minimum 500,000 packets per second (pps) forwarding rate
Protocol Compliance	IPv4, IPv6, TCP/IP, UDP, ICMP, OSPF, EIGRP, BGP, RIP, NAT, DHCP, VLAN, ACL, SNMP, RFC 1812

3. Wireless Access Points

Specification	Value
Technology	IEEE 802.11ax (Wi-Fi 6), dual band (2.4 GHz and 5 GHz)
Protocol Compliance	IEEE 802.11a/b/g/n/ac/ax, WPA2/WPA3, 802.11r/k/v, 802.1Q VLAN, OFDMA, MU-MIMO
PoE Support	IEEE 802.3af/at (PoE/PoE+)
Throughput	Up to 4.8 Gbps aggregate
Security	WPA2-Enterprise, WPA3-Enterprise, 802.1X, MAC filtering, rogue AP detection

4. UPS Power

Specification	Value
Capacity	1500 VA / 900 W per equipment rack
Type	Line-interactive UPS
Input Voltage	220V ± 10%
Output Voltage	220V ± 10% (pure sine wave)
Battery Type	Sealed Lead Acid
Backup Time	Minimum 10 minutes at 50% load
Features	Automatic Voltage Regulation (AVR), surge protection, LCD status display, USB monitoring interface

Cost Estimate

Component	Model / Type	Qty	Unit Price (LKR)	Total (LKR)
Router	TP-Link ER7206 / MikroTik hEX S (Gigabit + Firewall)	1	71,214	71,214
Core Switch (PoE)	TP-Link TL-SG2428P JetStream (24 PoE+ + 4 SFP)	1	91,960	91,960
Access Points	TP-Link EAP245 N1750 (Wi-Fi 5, Ceiling Mount)	2	41,000	82,000
Patch Panel (UTP)	Orange ONPP-U60324 (24-Port Cat6)	1	33,450	33,450
Fiber Patch Panel	8-Port SC (generic/local)	1	2,000	2,000
UTP Cable	GV-TECH Full Copper 100m UTP Roll	3	10,950	32,850
Fiber Cable	Armored Duplex LC/LC (Multimode, 1m)	10	895	8,950
Wall Outlets	Kevilton Modular Silver CAT6 Socket	16	4,950	79,200
Conduit Pipes (40mm)	Polycrome	4	2,170	8,680
Trunking (100x50 mm)	Polycon uPVC	3	4,395	13,185
Equipment Rack	12U Floor-Standing Network Rack	1	25,450	25,450
UPS	Eaton 5E 1500VA USB Line-Interactive	1	42,000	42,000
Installation & Configuration	Cabling, patching, switch & AP setup	–	–	25,000
Miscellaneous Items	Labels, cable ties, testing tools, patch cords	–	–	7,000
Total Estimated Cost				**522,939**

SECTION II - Simulation

1. Introduction

This simulation demonstrates the practical implementation and validation of a two-floor office network designed to support multiple departments and user categories. The network is logically segmented using Virtual LANs (VLANs) to separate staff with different usage requirements and guest users, improving security, performance, and manageability.

Inter-VLAN communication is enabled using a router-on-a-stick configuration, where a single router interface is subdivided into multiple sub interfaces to route traffic between VLANs. Dynamic Host Configuration Protocol (DHCP) is implemented to automatically assign IP addresses, default gateways, and DNS settings to end devices within each VLAN, reducing manual configuration and administrative overhead.

Domain Name System (DNS) services are configured to allow internal users to resolve domain names, while Network Address Translation (NAT) is used to provide controlled Internet access to all VLANs through a single public-facing interface. Access Control Lists (ACLs) are applied to enforce security policies, particularly restricting guest VLAN devices from accessing internal staff networks while still allowing essential services such as DHCP, DNS, and Internet connectivity.

Overall, this simulation serves as a functional proof-of-concept for the proposed physical network design, demonstrating correct VLAN segmentation, secure traffic control, and reliable network services in a small-to-medium enterprise environment.

2. Network Design Overview

2.1 Physical Network Layout

The physical network design represents the structural deployment of networking devices across a two-floor office environment. The topology follows a hierarchical model consisting of a core routing device and access-layer switches distributed across each floor.

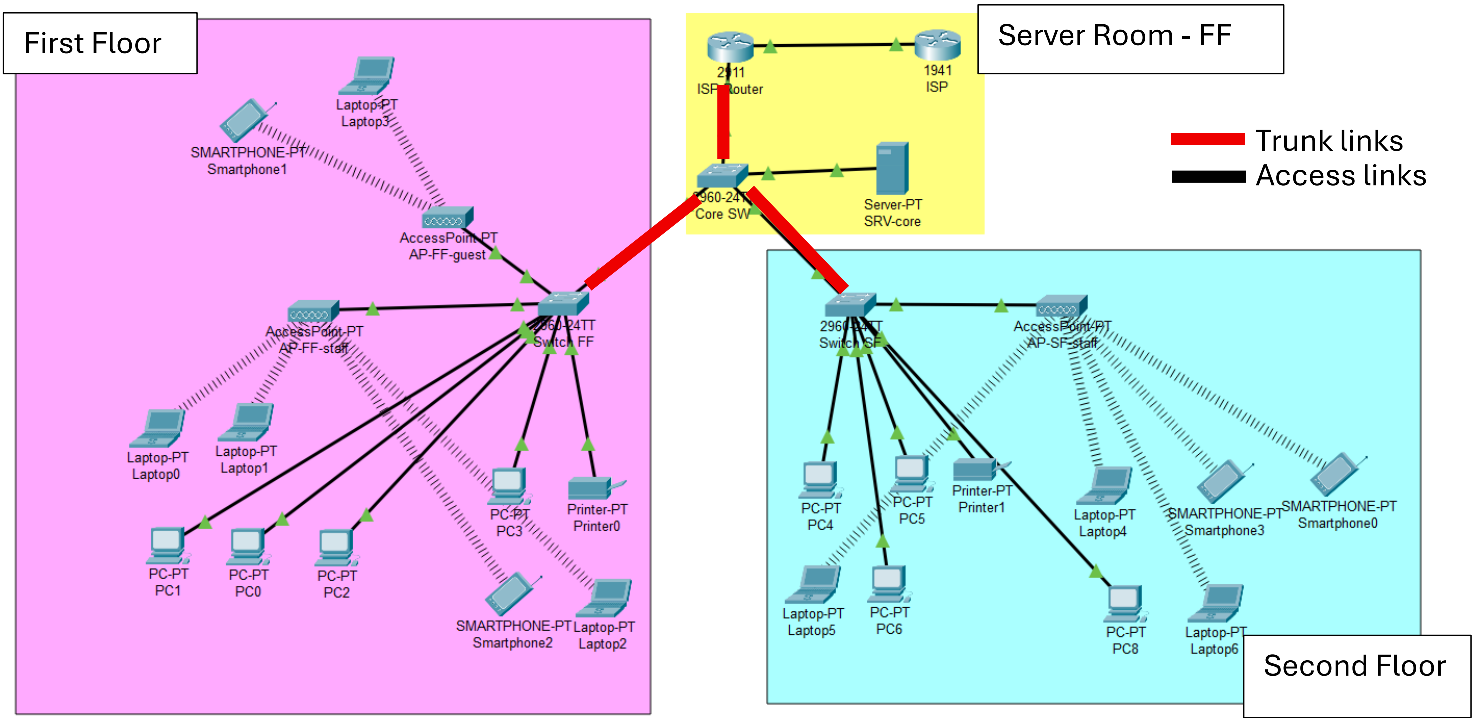
The infrastructure includes:

- One router providing inter-VLAN routing and WAN connectivity
- A core switch acting as the central aggregation point
- Access switches deployed on the First Floor (FF) and Second Floor (SF)
- Wireless access points installed to provide Wi-Fi coverage
- Wired end devices including desktop PCs and printers

- Wireless laptops connected via access points
- A server connected within the internal network
- A simulated ISP cloud representing external Internet connectivity

The core switch connects to the router via a trunk link, enabling multiple VLANs to traverse a single physical interface. Access switches on each floor connect to the core switch through trunk ports to maintain VLAN consistency across the network.

This physical segmentation ensures organized cabling, scalability, and structured device placement across both floors.



Physical topology of the two-floor office network implemented in Cisco Packet Tracer

2.2 Logical Network Design

The logical network design defines how traffic is segmented and controlled within the infrastructure. Virtual LANs (VLANs) are implemented to logically separate departments into distinct broadcast domains while sharing the same physical switching infrastructure.

The network is segmented as follows:

- **VLAN 10 – Staff Low Usage**
  Assigned to Sales, Administration, and Conference Room devices.

- **VLAN 20 – IT Department**
  Dedicated to IT personnel requiring full internal access for system management and troubleshooting.

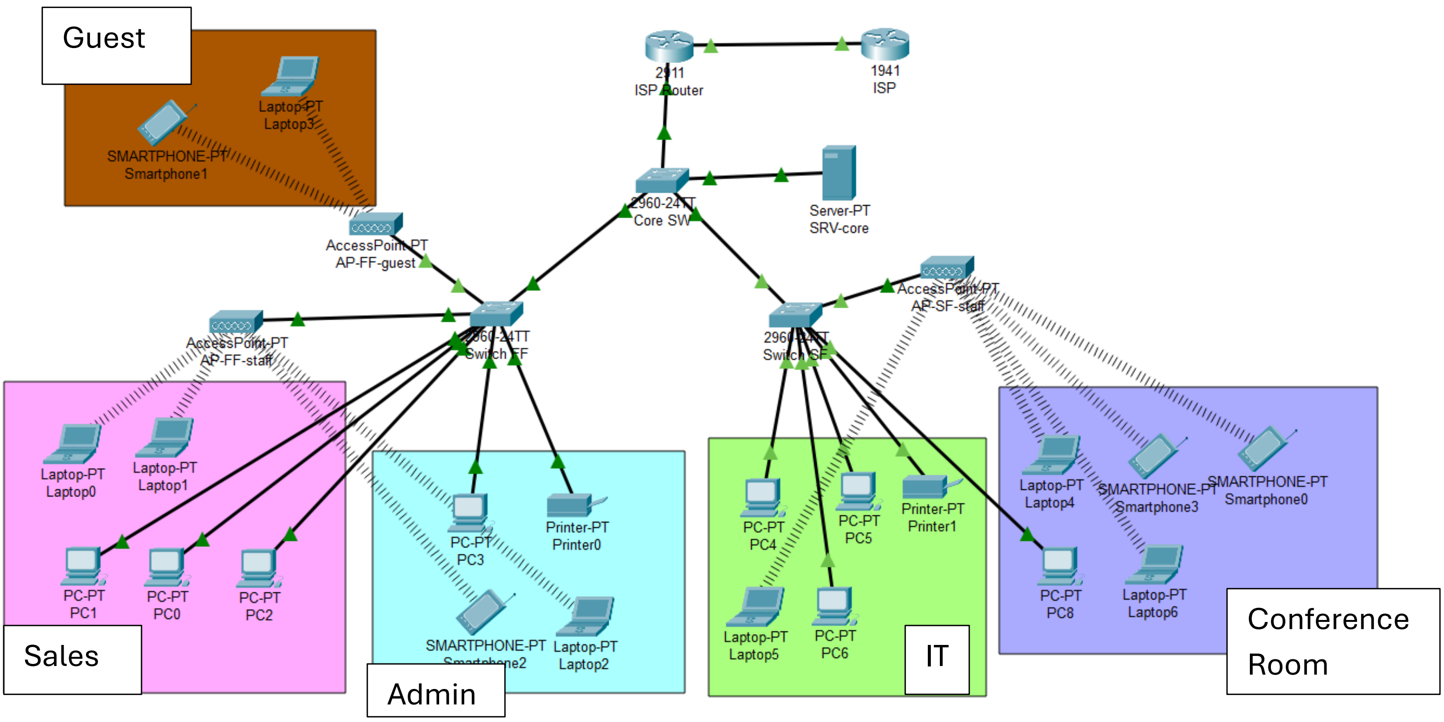
- **VLAN 30 – Guest Network**
  Isolated network providing Internet access only, with restricted access to internal staff resources.

Each VLAN is assigned its own IP subnet and default gateway on the router. Inter-VLAN communication is controlled at the routing layer, allowing centralized traffic management and policy enforcement.

In the logical diagram, VLANs are visually separated using colored annotations to demonstrate broadcast domain isolation while maintaining centralized routing through the router.

This design improves:

- Security through segmentation
- Broadcast traffic containment
- Administrative control at the routing layer
- Scalability for future departmental expansion



3. Device Configuration

3.1 Switch Configuration

VLANs 10, 20, and 30 were created on the core and access switches. Access ports were assigned to their respective VLANs, and uplink interfaces were configured as trunk ports to carry tagged traffic between switches and the router.

```
Switch#
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Gig0/1      on           802.1q         trunking      1
Gig0/2      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005
Gig0/1      1-1005
Gig0/2      1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20,30
Gig0/1      1,10,20,30
Gig0/2      1,10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30
Gig0/1      1,10,20,30
Gig0/2      1,10,20,30

Switch#
```

Core Switch configuration

```
Switch>enable
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   STAFF                            active    Fa0/6, Fa0/7
20   IT                               active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
30   GUEST                            active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Second floor switch configuration

3.2 Router Configuration

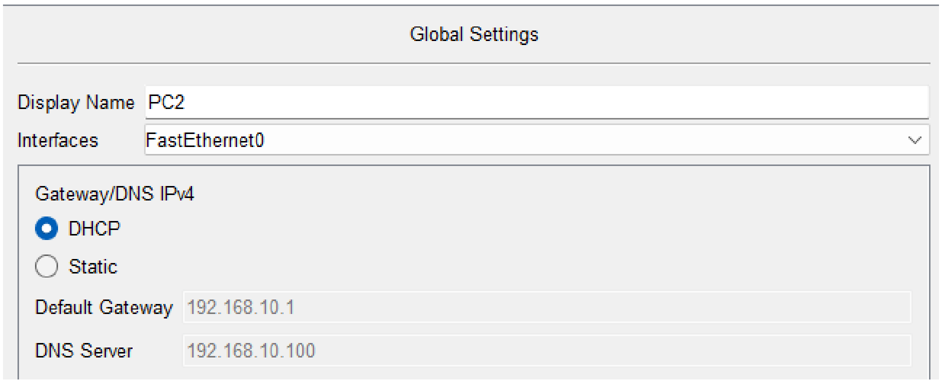
Inter-VLAN routing was implemented using router-on-a-stick. Subinterfaces were created for each VLAN, with corresponding gateway IP addresses.

NAT was configured on the WAN interface to enable Internet access for internal private networks.

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     200.1.1.2       YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

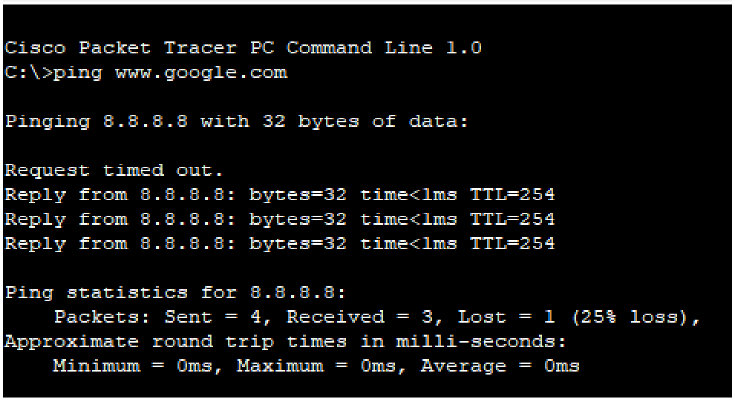
3.3 DHCP Configuration

Router-based DHCP pools were configured for each VLAN subnet. Exclusion ranges were defined for infrastructure devices. Clients automatically receive IP address, gateway, and DNS settings.



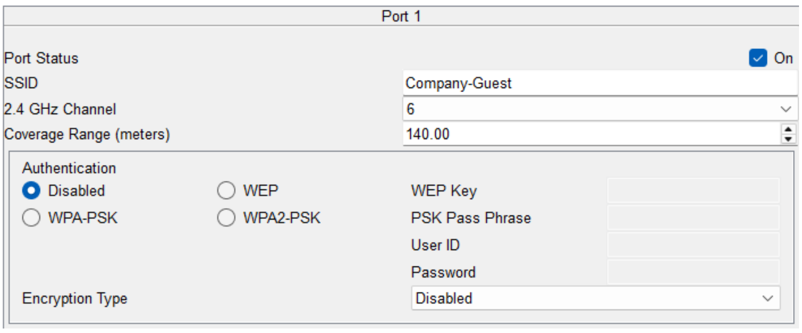
3.4 DNS Verification

Clients use DNS server **192.168.10.100**, distributed via DHCP. Successful domain resolution was verified through external domain ping testing.



3.5 Wireless Access Points

Wireless access points were configured with dedicated SSIDs to provide wireless connectivity for users. VLAN segmentation for wireless clients is enforced at the connected switch ports, ensuring traffic is assigned to the appropriate VLAN based on network design.



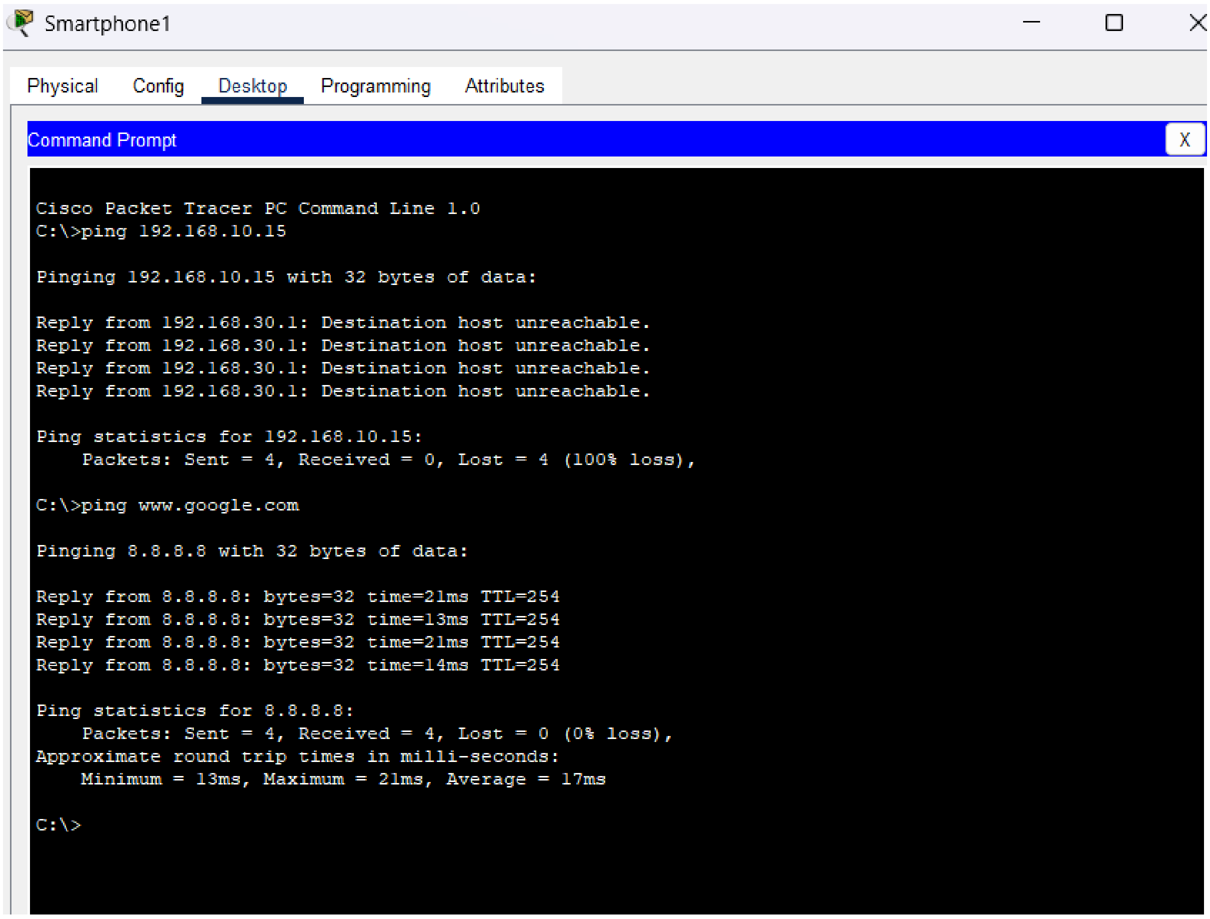
4. Access Control

To secure the guest VLAN (VLAN 30) while still allowing Internet access, access control lists (ACLs) were applied on the router sub interface for VLAN 30. The ACLs enforce the following policies:

- **Deny traffic** from VLAN 30 to internal VLANs (VLAN 10 & VLAN 20) to prevent unauthorized access.
- **Allow DHCP** so guest devices can receive IP addresses from the router.
- **Allow DNS** to the DNS server (192.168.10.100) so guests can resolve domain names.
- **Permit all other traffic** to the Internet.

This setup ensures that guests can access online resources without compromising internal network security.

The following image shows a ping test done from a device in VLAN 30.



5. Observations and Conclusion

The simulation successfully implemented the planned two-floor office network. VLANs were correctly segmented according to department and usage, and inter-VLAN routing via router-on-a-stick allowed controlled communication between VLANs. DHCP on the router correctly assigned IP addresses to devices, and DNS resolution worked for authorized VLANs. NAT provided Internet access for all internal users.

ACLs on the guest VLAN effectively restricted internal access while allowing DHCP and DNS services, ensuring security without impacting Internet connectivity. Troubleshooting steps included verifying VLAN assignments, trunk ports, router sub interfaces, ACL rules, and NAT configuration.

Overall, the simulation demonstrates practical understanding of VLAN segregation, router-on-a-stick configuration, DHCP and DNS operation across VLANs, and ACL-based network security.